BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz - 1 Semester: Summer2025
Duration: 25 minutes Full Marks: 15

CSE 340: Computer Architecture

Name: Solution	ID:	Section:

1. It takes 10s to run the following instruction set in computer A.

6 instructions
Add ……………..
Sub ……………..
Add ……………..
Addi ……………..
Add ……………..
Sub ……………..

Find the MIPS of computer A? [3]
Answer:

10s — 6 ins
1s — $\frac{6}{10}$ ins

10,00000 — 1M
1 — $\frac{1}{10^6}$ M
$\frac{6}{10}$ — $\frac{6}{10} \times \frac{1}{10^6} = 6 \times 10^{-7}$ M $\Rightarrow 6 \times 10^{-7}$ MIPS

OR

$$\text{MIPS} = \frac{\text{Ins. Count}}{\text{Execution time} \times 10^6}$$
$$= \frac{6}{10 \times 10^6}$$
$$= 6 \times 10^{-7}$$

2. Suppose you are working in a chipset manufacturing company, and while analyzing a newly designed IC, you observe that the internal architecture and implementation details of the IC are hidden from end-users and developers. This allows the chip to be upgraded or optimized internally without affecting external usage or compatibility.
Write the name of the relevant idea from the 7 Great Ideas that is applied in the given scenario [2]
Answer:

Use $\boxed{\text{abstraction}}$ to simplify design.

3. Consider the execution of Program A, which consists solely of add instructions, on two different computer systems: a reference computer and your computer. On the reference computer, running Program A requires a total of 2×10^9 instructions. The reference computer operates with a clock rate of 3 GHz and has a CPI of 3.
On your computer, the total number of clock cycles required to execute Program A is 7×10^9. Your computer operates with a clock rate of 2.5 GHz, and it is observed that the execution time to run Program A on your computer is 5 seconds.

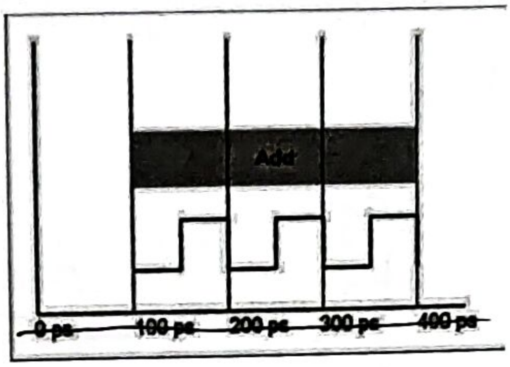

Figure - 1 : Duration of one ADD instruction in your Computer ~~A~~

a) Find the reference time. [3]
b) What is the CPI of Add instruction for your Computer ~~A~~? [2]
c) Find the instruction count of your computer. [2]
d) Find the spec ratio. [2]
e) Comment about the performance of your computer based on the spec ratio. [1]
Answer:

a) Reference time is the execution time of the reference computer.

For ref. com., Instruction count = 2×10^9

Clock Rate = 3GHz = 3×10^9 Hz

CPI = 3

$$\therefore \text{Ref. time} = \frac{IC \times C.P.I}{C.R}$$

$$= \frac{2 \times 10^9 \times \cancel{3}}{\cancel{3 \times 10^9}}$$

$$= 2s$$

b) CPI = 3

c) C.C = Instruction count × CPI

$$\Rightarrow I.C = \frac{C.C}{CPI}$$

$$= \frac{7 \times 10^9}{3}$$

$$= 2.33 \times 10^9$$

OR

$$CPU\ time = \frac{IC \times CPI}{Rate}$$

$$\Rightarrow 5 = \frac{IC \times 3}{2.5 \times 10^9}$$

$$\Rightarrow IC = \frac{5 \times 2.5 \times 10^9}{3}$$

$$= 4.167 \times 10^9$$

well the answers should have been same but as I have set the CPU time randomly to 5s, Hence the different answers.
If you used a correct approach, you will get full marks.

d) $$Spec\ Ratio = \frac{Ref.\ time}{execution\ time}$$

$$= \frac{2}{5} = 0.4$$

e) 0.4 < 1; so, my computer is slower

or, Prog. A runs slower in my computer than the ref. computer.